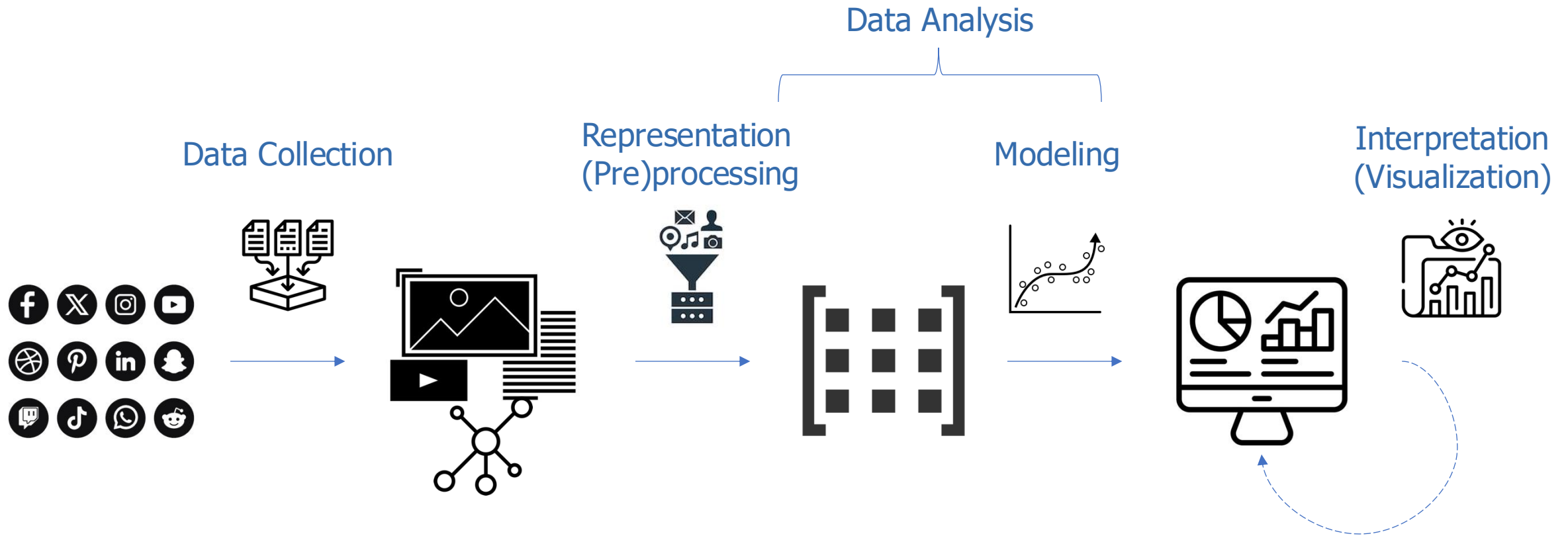


AS Module: Scrape Web Data

Social media analytics process

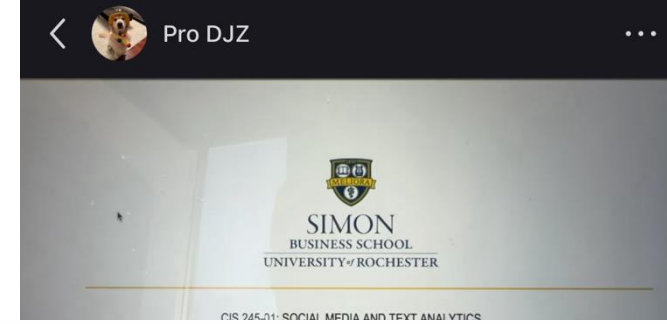
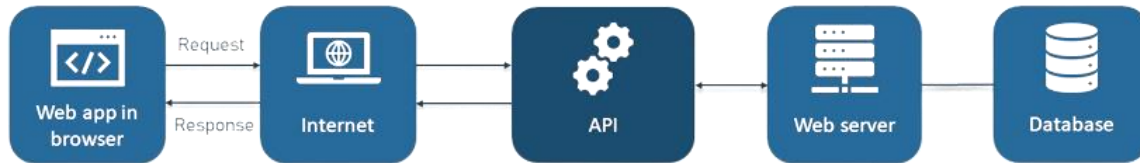


Agenda for AS Module

- Web structure
- Scrape one feature on one page
- Scrape hierarchical (more than one) features
- Scrape with pagination (across pages)
- Scrape hierarchical pages

Scrape web data

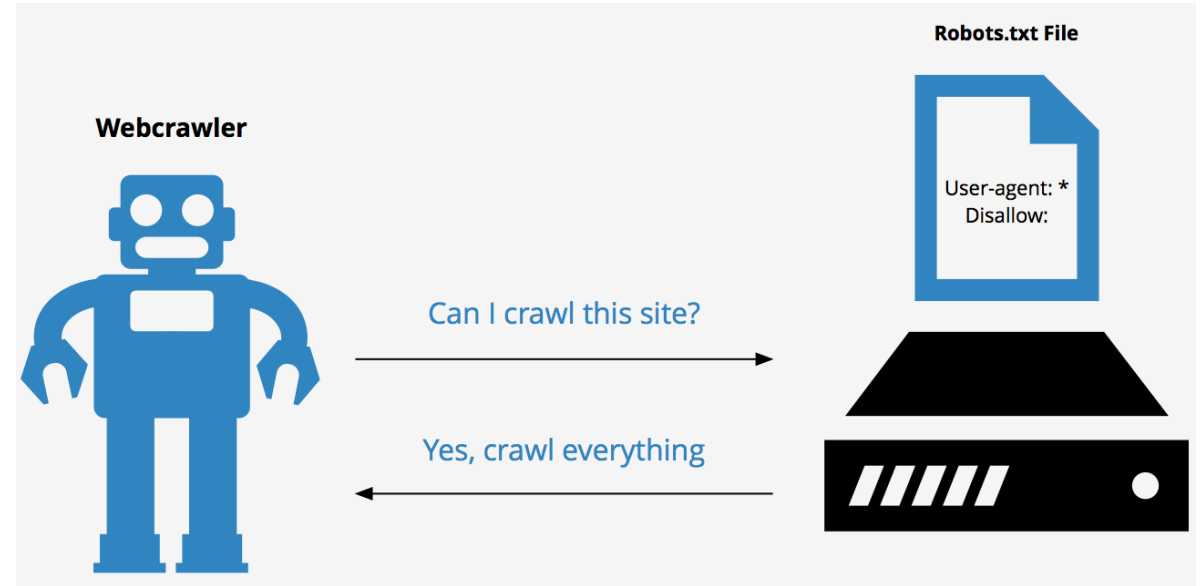
- Why you want to scrape data?
 - Unstructured data vs. structured data
 - Not owned by most companies
 - There was no API or limited API offering
 - An invincible tool



Don't scrape on them!!

When you cannot scrape data?

- Intellectual property owned and violation
- Terms of services violation
- Robots Exclusion Protocol (REP) robots.txt
- Disregard robots.txt? Yes but no!!
- Check with ChatGPT



How to scrape data?

- How would you scrape data manually?
- Step 1: visit the webpage **URL**
- Step 2: select the data **Vision**
- What about using computer?
- Nothing different but using a robot/crawler/spider
- Automate the scrape actions
- Step 1: **browse: request()** **URL**
- Step 2: **selector: response() and extract()** **? Scripts/code**

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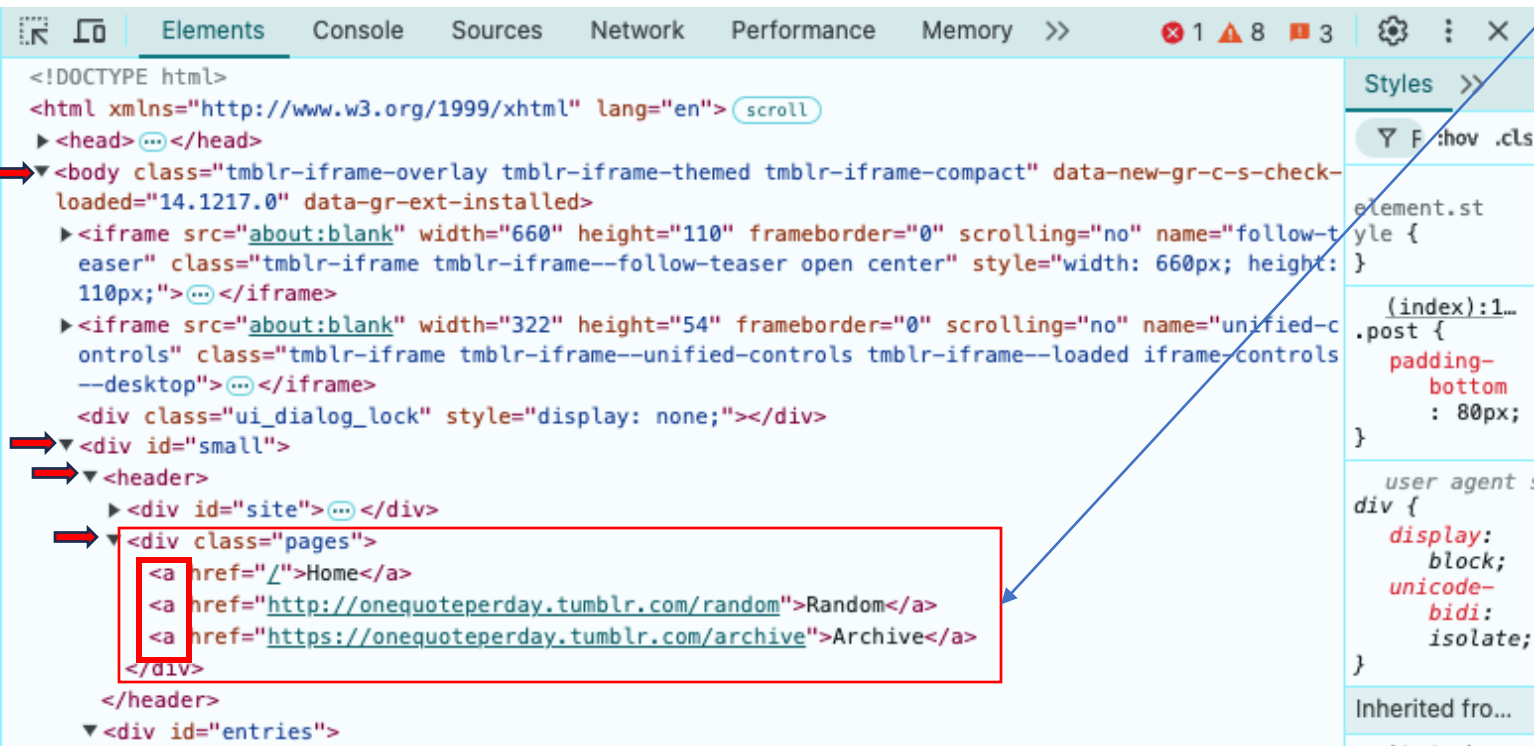
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```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml" lang="en">
  <head>
    <body class="tmblr-iframe-overlay tmblr-iframe-themed tmblr-iframe-compact" data-new-gr-c-s-check-loaded="14.1217.0" data-gr-ext-installed>
      <iframe src="about:blank" width="660" height="110" frameborder="0" scrolling="no" name="follow-teaser" class="tmblr-iframe tmblr-iframe--follow-teaser open center" style="width: 660px; height: 110px;"></iframe>
      <iframe src="about:blank" width="322" height="54" frameborder="0" scrolling="no" name="unified-controls" class="tmblr-iframe tmblr-iframe--unified-controls tmblr-iframe--loaded iframe-controls--desktop"></iframe>
      <div class="ui_dialog_lock" style="display: none;"></div>
      <div id="small">
        <header>
          <div id="site">
            <div class="pages">
              <a href="/">Home</a>
              <a href="http://onequoteperday.tumblr.com/random">Random</a>
              <a href="https://onequoteperday.tumblr.com/archive">Archive</a>
            </div>
          </header>
          <div id="entries">
            <div class="post text">
              <article>
                <h2></h2>
                <p>
                  "...and thanks for all the fish. "
                </p>
                <p>
                  "Today's quote was the last one – after 12 years (and some change), I am closing this blog down. As in, there won't be any new posts, but you can still hit the "
                  <a href="http://onequoteperday.tumblr.com/random">Random</a>
                  " button each day to read one (or more) of the 4395 quotes collected on this blog. There's definitely a bunch of duplicates but it's the thought that counts."
                </p>
                <p></p>
              </article>
              <div class="post_footer">
                <!-- Post Footer -->
              </div>
              <div class="post quote">
                <span>"A book of quotations ... can never be complete."</span>
                <div class="source">Robert M. Hamilton</div>
                <div class="tags"></div>
                <div class="post_footer">
                <!-- Post Footer -->
              </div>
            </div>
          </div>
        </div>
      </div>
    </body>
  </html>
```

Inspect web scripts



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      <div id="entries">
        <div class="post text">
        <div class="post quote">
          <span>'A book of quotations ... can never be complete.'</span>
          <div class="source">Robert M. Hamilton</div>
          <div class="tags">
          <div class="post_footer">
            <!-- Post Footer -->
          </div>
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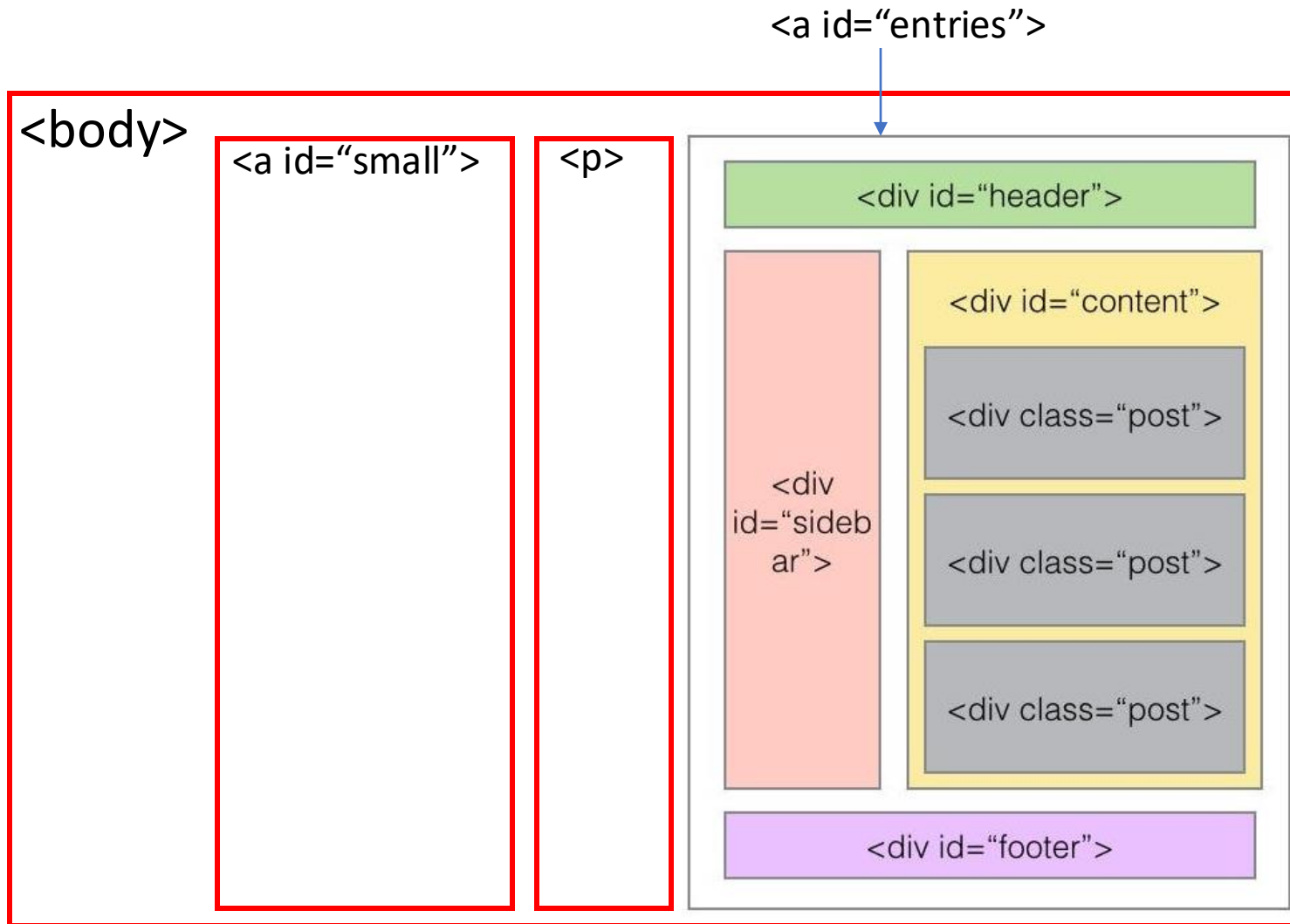
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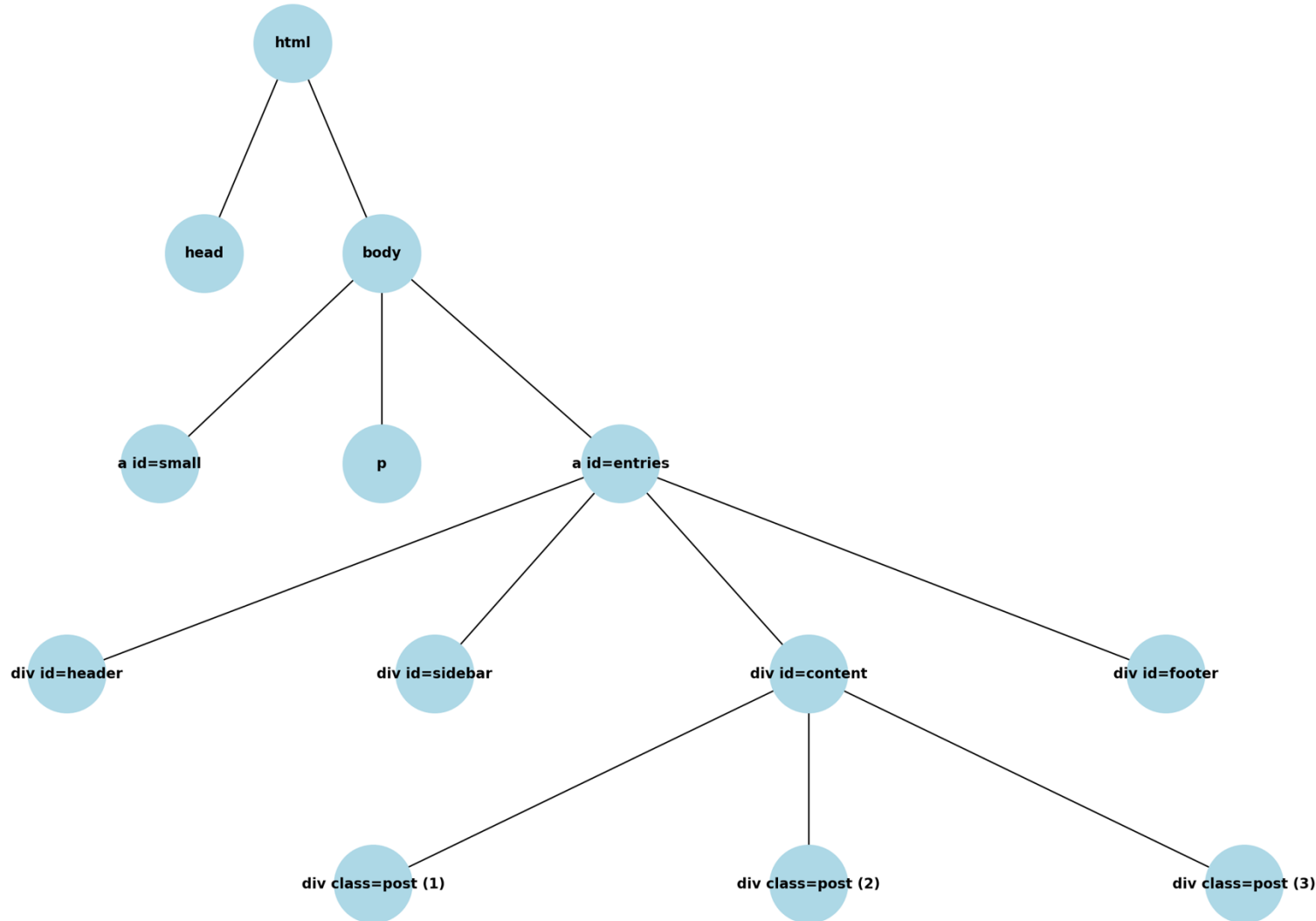
HTML Structure



```
<html>
  <head>
  <body>
    ➤ <a id="small">
    ➤ <p>
    ➤ <a id="entries">
      ➤ <div id="header">
      ➤ <div id="sidebar">
      ➤ <div id="content">
        ➤ <div class="post">
        ➤ <div class="post">
        ➤ <div class="post">
      ➤ <div id="footer">
```

Hierarchical Tree

/html /body /a[@id="entries"] /div[@id="content"] /div[@class="post"]



<html>

<head>

<body>

➤

➤ <p>

A unique <tag>

➤

➤ <div id="header">

➤ <div id="sidebar">

➤ <div id="content">

➤ <div class="post">

➤ <div class="post">

➤ <div class="post">

➤ <div id="footer">

A unique group of <tags>

Specify xpath

- // start from the beginning <html
- / go to the next level tag, or attribute and text within the final tag
- @ select an attribute, e.g., "id"=, "class"=, "href"=
- text() select the text in a tag
- [] impose a selection condition on a tag

```
➤ <html>
  ➤ <header>
  ➤ <body>
    ➤ <span class="a">
      ➤ <span>
        ➤ <div class="body">
          ➤ <span>"Please locate me also"<\span>
        ➤ <div>
          ➤ <div class="source">"Please locate me"<\div>
          ➤ <div class="url" href="http://google.com/">
          ➤ <span class="b">
          ➤ <span>
```

- //body/div[@class="source"]/text()
- //body/span[@class="a"]/div[@class="body"]/span/text()
- //body/div[@class="url"]/@href

Full path vs. a shortcut

- Can skip all the path specification before the last tag with a class or id attribute
- This is because a tag with class or id attribute can be (uniquely) identifiable
- Path after the last tag with class or id attribute cannot be suppressed

- `//div[@class="source"]/text()`
- `//div[@class="body"]/span/text()`
- `//div[@class="url"]/@href`

```
➤ <html>
  ➤ <header>
  ➤ <body>
    ➤ <span class="a">
      ➤ <span>
        ➤ <div class="body">
          ➤ <span>"Please locate me also"<\span>
        ➤ <div>
          ➤ <div class="source"> "Please locate me"<\div>
          ➤ <div class="url" href="http://google.com/">
          ➤ <span class="b">
          ➤ <span>
```

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